

12 A planet has an average density 4950 kg m^{-3} . Given that the acceleration of free fall near the surface of the planet is $0.9 g$, where g is the acceleration of free fall near Earth's surface, the radius of the planet is approximately

A 260 km

B 2500 km

C 2800 km

D 6400 km

Ans: D

$$g = \frac{GM}{R^2} \rightarrow g = \frac{G}{R^2} \frac{4}{3} \pi R^3 \rho \rightarrow R = \frac{3g}{4\pi G \rho}$$

$$R = \frac{3(0.9 \times 9.81)}{4\pi(6.67 \times 10^{-11})(4950)} = 6400 \text{ km}$$

13 Two satellites A and B orbiting around Earth have the same kinetic energy. Satellite A has a